

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Stochastics Recap

Oct 29, 2025

Welcome to the lecture!

Prof. Dr.-Ing. Björn Giesler

Technische Hochschule Ingolstadt

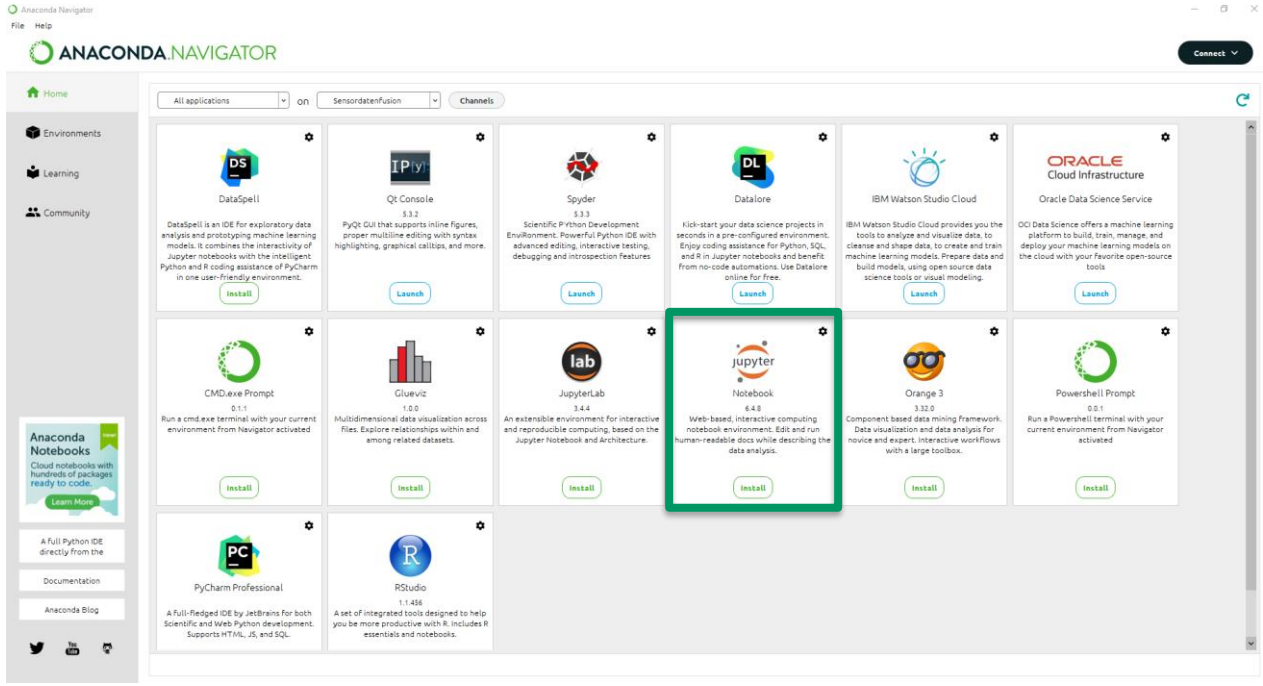
AIMotion, AI and Sensor Data Fusion

Email: bjoern.giesler@thi.de

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

Software: Python/Anaconda

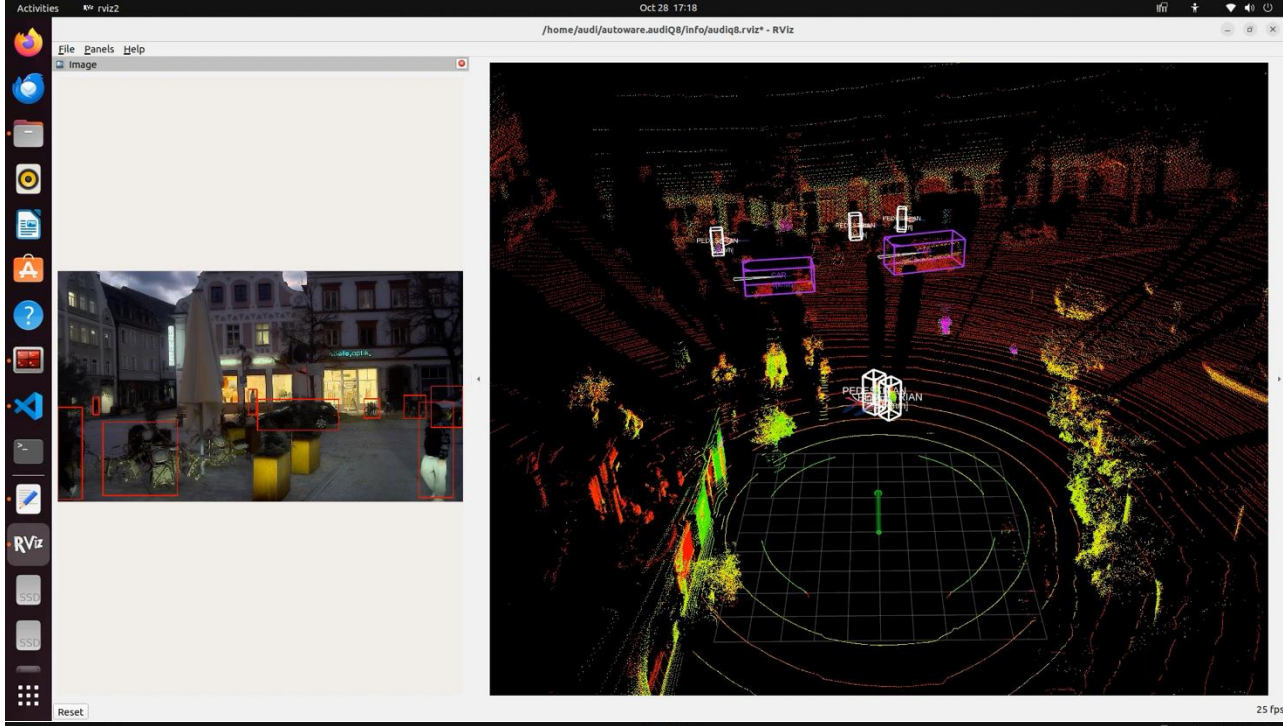
- For the ML examples we will use today, will use the Jupyter Notebook environment.
- It is recommended you install the Anaconda Python environment and distribution, that installs Jupyter as one of its environments.
- You can find it at https://www.anaconda.com/.

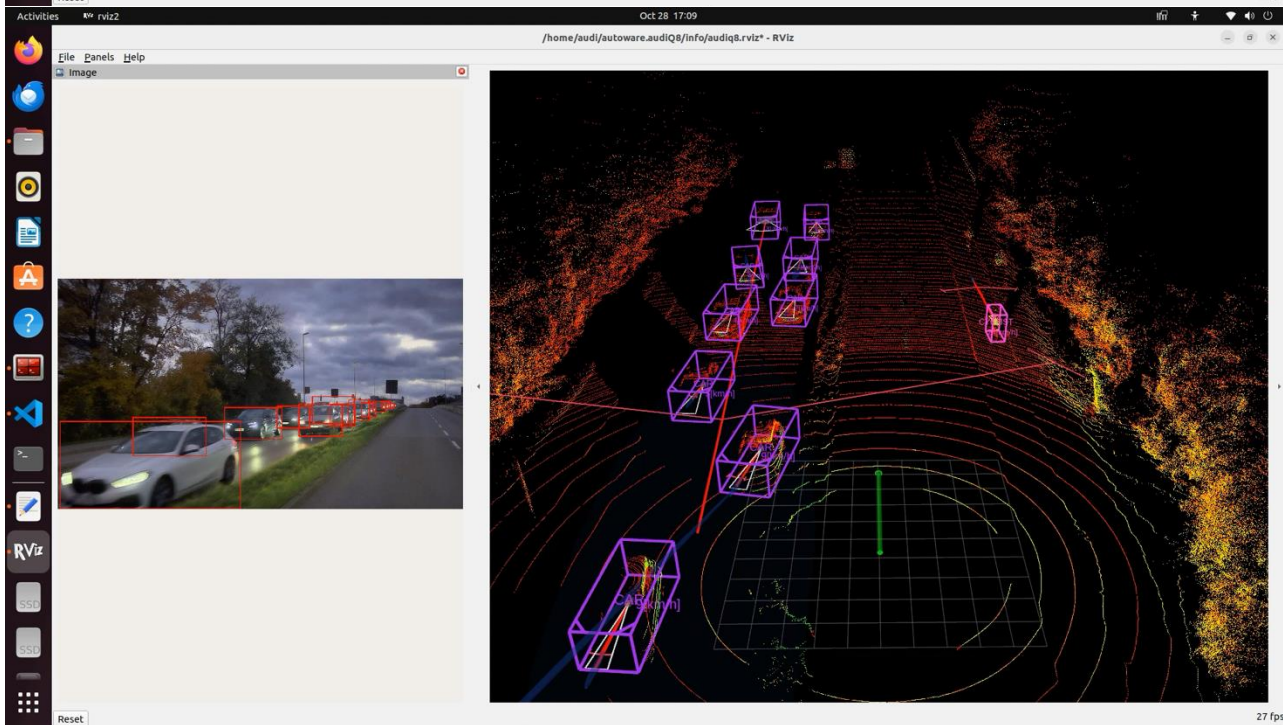


Sensor Networks and Sensor Fusion

Updates @ the SDF Group at AIMotion

- **The SDF team is part of the AIMotion Institute.**
- **AIMotion is equipped with 3 sensor fusion vehicles and currently building up a Drive By Wire autonomous vehicle.**
- **Progress and Goals:**
  - Get basic multi-sensor perception running until end of 2025
  - Work on safety fallbacks, fusion and planning/control starting 2026
- **Master theses and similar work:**
  - Cam/Lidar/Radar Early/Mid Level Fusion (~2)
  - Geometric / non-AI Detection and Fusion (~2)
  - Fusion-to-Wheels Planning / Control (~2)
- **If you‘re interested please give me a shout at bjoern.giesler@thi.de!**





A Glimpse of State Modeling

Let‘s assume we want to determine the motion of an object moving in a straight line.

Let‘s also assume we want to model its position, velocity, and acceleration.

- **The object‘s system state at time *t* can be expressed as** $\mathbf{x}_t = \begin{bmatrix} s_t \\ v_t \\ a_t \end{bmatrix}$.

- **As time progresses, the object‘s state changes according to a model:**

$$\left.\begin{aligned} s_{t+1} &= s_t + \Delta t \cdot v_t + \frac{1}{2}(\Delta t)^2 \cdot a_t \\ v_{t+1} &= v_t + \Delta t \cdot a_t \\ a_{t+1} &= a_t \end{aligned}\right\} \quad \mathbf{x}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t \quad \textbf{with} \quad \mathbf{A} = \begin{bmatrix} 1 & \Delta t & (\Delta t)^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}.$$



Let‘s also assume we measure the object with a sensor, let‘s say a radar, that can measure distance and velocity but not acceleration:

$$\hat{\mathbf{z}}_t = \begin{bmatrix} \hat{s}_t \\ \hat{v}_t \end{bmatrix}$$

We can express an estimate of what *z* should look like by creating a measurement matrix H:

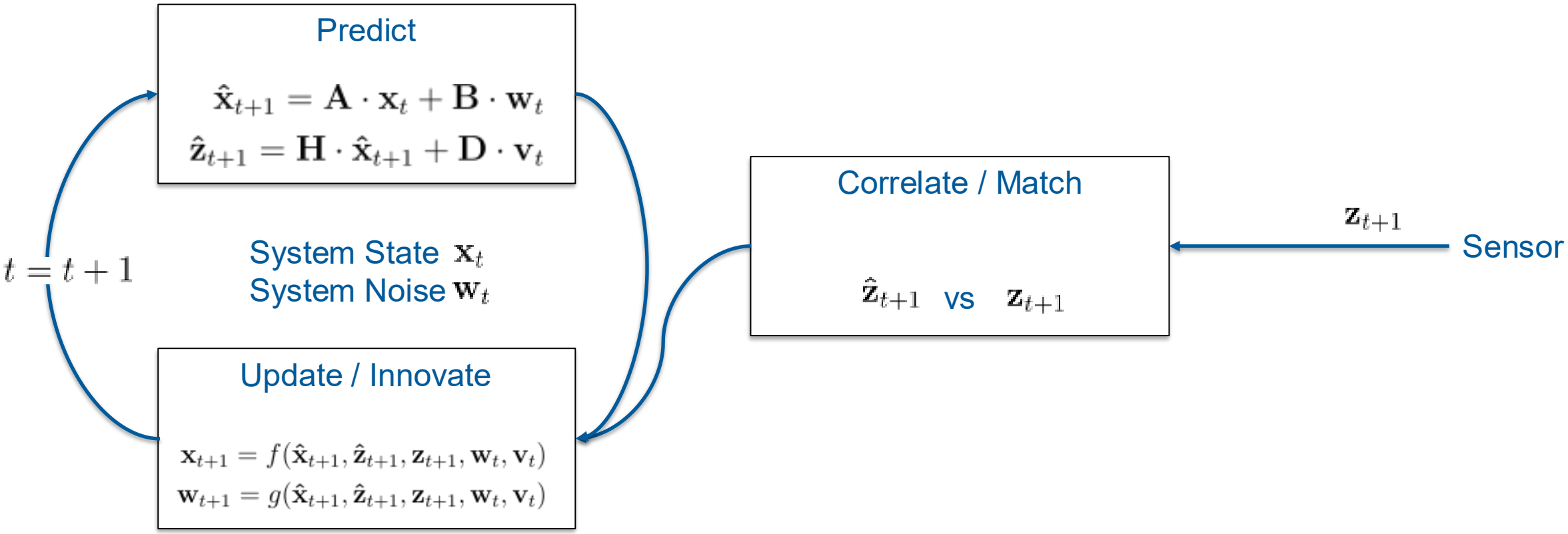
$$\mathbf{z}_t = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} s_t \\ v_t \\ a_t \end{bmatrix} = \mathbf{H}\mathbf{x}_t$$

We expect to make an error both in measuring (because of noise and other residual errors) and in updating the model (because of inaccuracies in the model itself, and previous errors).

So we can write our estimates for the new state and measurement as

$$\hat{\mathbf{x}}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t + \mathbf{B} \cdot \mathbf{w}_t$$
$$\hat{\mathbf{z}}_{t+1} = \mathbf{H} \cdot \hat{\mathbf{x}}_{t+1} + \mathbf{D} \cdot \mathbf{v}_t$$

with $\mathbf{B} \cdot \mathbf{w}_t$ **to denote model uncertainty and**
$\mathbf{D} \cdot \mathbf{v}_t$ **to denote measurement uncertainty.**


Predict
$\hat{\mathbf{x}}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t + \mathbf{B} \cdot \mathbf{w}_t$
$\hat{\mathbf{z}}_{t+1} = \mathbf{H} \cdot \hat{\mathbf{x}}_{t+1} + \mathbf{D} \cdot \mathbf{v}_t$
$t = t + 1$
System State $\mathbf{x}_t$
System Noise $\mathbf{w}_t$
Update / Innovate
$\mathbf{x}_{t+1} = f(\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{z}}_{t+1}, \mathbf{z}_{t+1}, \mathbf{w}_t, \mathbf{v}_t)$
$\mathbf{w}_{t+1} = g(\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{z}}_{t+1}, \mathbf{z}_{t+1}, \mathbf{w}_t, \mathbf{v}_t)$
Correlate / Match
$\hat{\mathbf{z}}_{t+1}$ vs $\mathbf{z}_{t+1}$
$\mathbf{z}_{t+1}$
Sensor

Random Variables

Example: Random Experiment

- Throwing a fair dice: Six possible outcomes ω_i from sample space

$$\Omega = \{"1", "2", "3", "4", "5", "6"\}$$

Random Variable / Realization of a Random Variable:

- A **function** $\mathrm{X}(\omega)$, which assigns a real number x to every outcome $\omega \in \Omega$, is called **random variable**, so we have

$$x = X(\omega)$$

- The real number x for one specific outcome is denoted as **realization of a random variable** $\mathrm{X}(\omega)$,

Example 1:

- Throwing a dice: Possible outcomes ω_i

$$\Omega = \{"1", "2", "3", "4", "5", "6"\}$$

- Number of points on the dice (realization) $x_i = X(\omega_i)$

Example 2:

- Throwing a dice : Oucomes ω_i are heads and tails
- Assign a real number to every outcome, i. e. head $\rightarrow 0$, tail $\rightarrow 1$, i.e. $X(\omega) \in \{0, 1\}$

Events

- Sample Set
  $$\Omega = \{\omega_1, \omega_2, \ldots, \omega_L\}$$
- Define an Event $A = \{a_1, a_2, \ldots, a_M\} \subseteq \Omega$ with $M \leq L$
- Certain event $A = \Omega$
- Impossible event $A = \emptyset$
- Complementary event A^c
- Disjoint events: $A \cap B = \emptyset$ (event A and B can not occur jointly)

Example: Throwing a dice

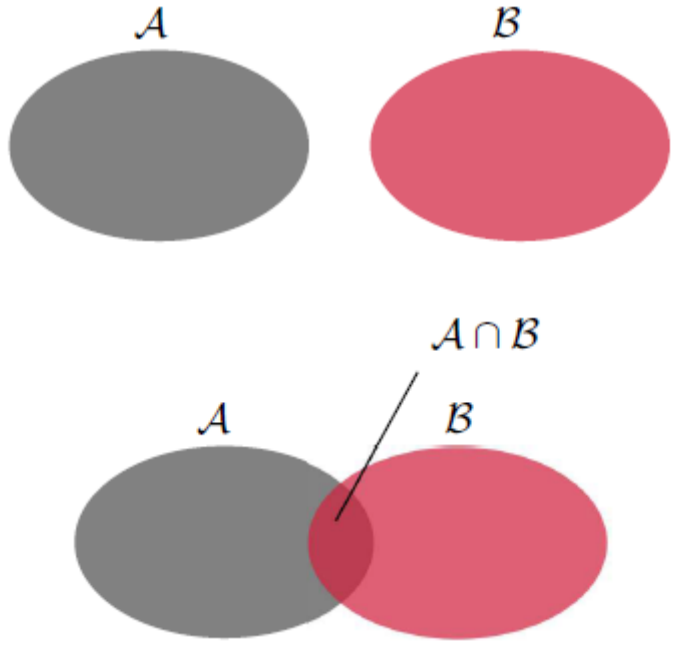
- Sample set of possible outcomes ω_i



- Define events:
  - Throwing a „3“, $A = \{3\}$
  - Throwing an even number $A = \{2, 4, 6\}$
- Complementary event for $A = \{2, 4, 6\}$ is $A^c = \{1, 3, 5\}$
- Disjoint event $A \cap B$ with $A = \{2, 4, 6\}$ and $B = \{7\}$

Events

- Sample Set
  $$\Omega = \{\omega_1, \omega_2, \ldots, \omega_L\}$$
- Define an Event $A = \{a_1, a_2, \ldots, a_M\} \subseteq \Omega$ with $M \leq L$
- Certain event $A = \Omega$
- Impossible event $A = \emptyset$
- Complementary event A^c
- Disjoint events: $A \cap B = \emptyset$ (event A and B can not occur jointly)



Disjont events (top) and non-disjoints events (bottom)

Frequency of Events and Probability

- Frequency $h(A)$ and probability $P(A)$

$$P(A) = \lim_{n\to\infty} h_n(A) \quad \text{with} \quad h_n(A) = \frac{\text{number of experiments with outcome A}}{n}$$

- **Axioms of Kolmogorov**
  - Probability for an event with an outcome between 0 and 1, i. e. it holds $0 \leq P(A) \leq 1$
  - The “certain event” (that is certain to occur) has $P(\Omega) = 1$
  - For disjoint events $A \cap B = \emptyset$ we have that $P(A \cup B) = P(A) + P(B)$

Conclusions:

- Probability of a complementary events A^c is $P(A^c) = 1 - P(A)$
- Probability of the impossible event is $P(\emptyset) = 0$
- Union of non-disjoint events $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Frequency of Events and Probability
- Frequency $h(A)$ and probability $P(A)$
- **Axioms of Kolmogorov**
  - Probability for an event with an outcome between 0 and 1, i. e. it holds $0 \leq P(A) \leq 1$
  - The “certain event” (that is certain to occur) has $P(\Omega) = 1$
  - For disjoint events $A \cap B = \emptyset$ we have that $P(A \cup B) = P(A) + P(B)$

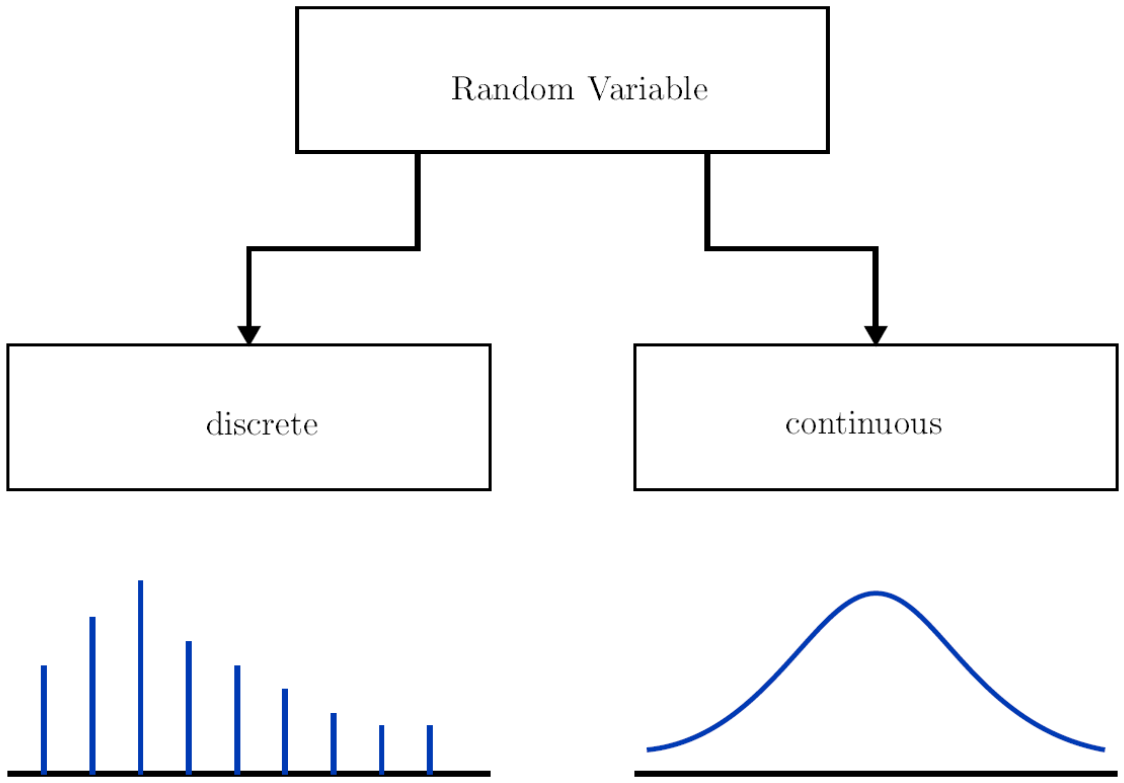
Conclusions:
- Probability of a complementary events A^c is $P(A^c) = 1 - P(A)$
- Probability of the impossible event is $P(\emptyset) = 0$
- Union of non-disjoint events $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Union of non-disjoint events

Probability Density Function

Discrete vs. Continuous random Variables

- Discrete random variable: Only discrete (Throwing dice/a coin)
  - Discrete probability density function
- Continuous random variable: Realization can take arbitrary values (Analogous output voltage of a sensor)
  - Continuous probability density function



Probability density functions of discrete and continuous random variables

Probability Density Function (PDF)

- Definition (for continuous random variables X) – compare Riemann integrals!

$$p_X(x) = \lim_{dx \to 0} \frac{P(x - dx < x \leq x + dx)}{dx}$$

- Therefore, probability that the realization x takes on values between a and b is computed by integration:

$$P(a < x \leq b) = \int_a^b p_X(x) dx$$

- The probability of the certain event

$$P(x \leq \infty) = \int_{-\infty}^{\infty} p_X(x) dx = 1$$

- Note that the probability of the **complementary event** $P(x = C) = 0$
- Please also note the relation to the **(cumulative) distribution function**

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} p_X(x) dx$$

Preliminary Remarks:

- Continuous space of possible events (before: discrete)
- Mathematically, discrete events can be described in terms of a continuous probability density function by means of the Dirac-distribution:

$$\delta(x) = \begin{cases} +\infty & \text{if} \quad x = 0 \\ 0 & \text{otherwise} \end{cases}$$

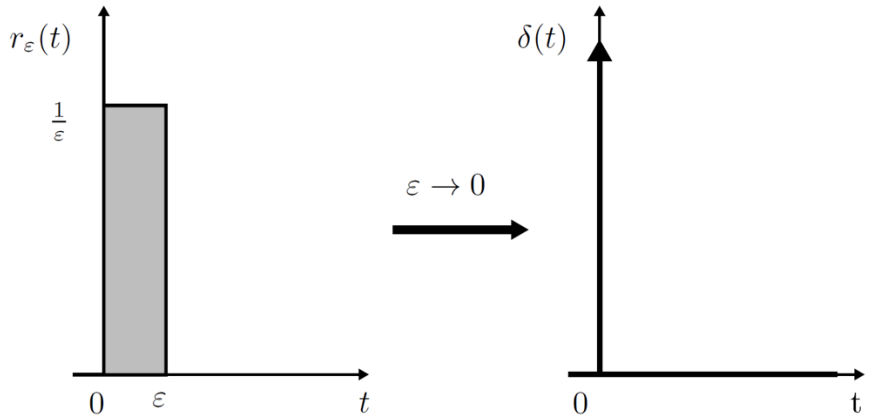
- Gating-characteristic: Due to

$$\int_{-\infty}^{\infty} \delta(t)dt = 1,$$

we have that

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$$

Definition of the Dirac-Function



- Rectangle-shaped function:

$$r_\epsilon(t) = \begin{cases} \frac{1}{\epsilon} & \text{if} \quad 0 \leq t \leq \epsilon \\ 0 & \text{otherwise} \end{cases}$$

- Limiting operation:

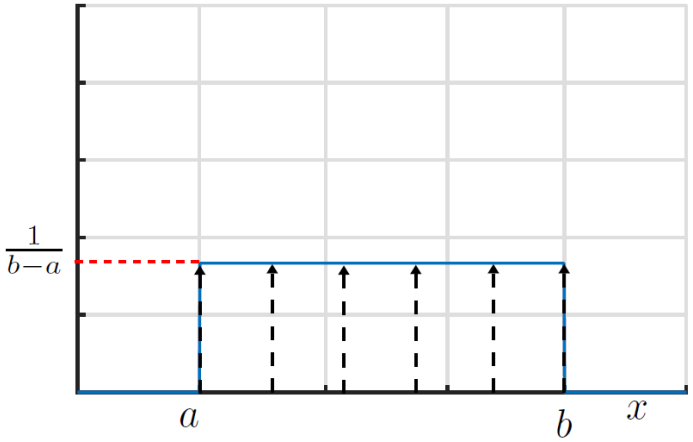
$$\delta(t) = \lim_{\varepsilon \to 0} r_\varepsilon(t)$$

Uniform Distribution:

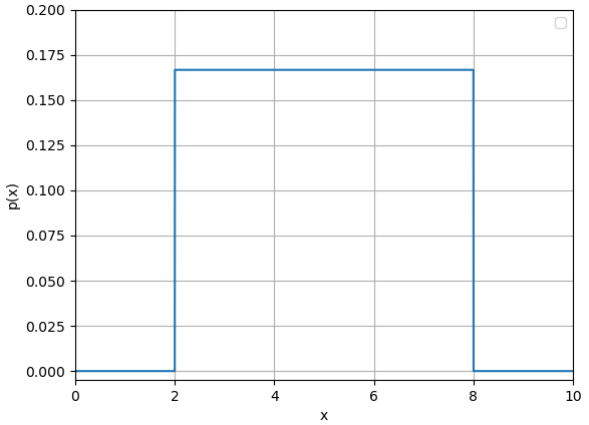
- The probability distribution of a uniformly distributed random variable is

$$p_X(x) = \begin{cases} \frac{1}{b-a} & \text{if} \quad a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

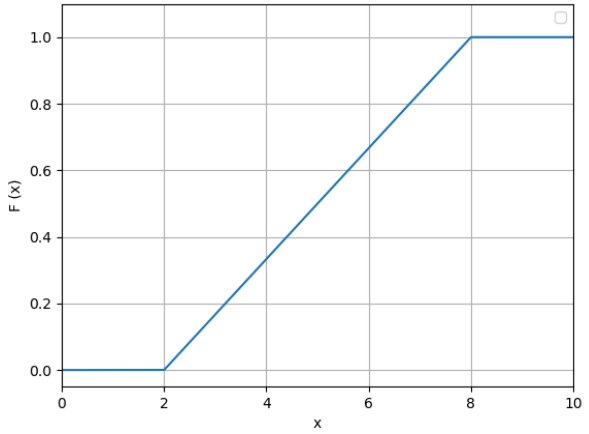
  - The discrete counterpart can be represented by means of a convolution with the Dirac distribution (e. g. throwing a dice)



Uniform distribution: Discrete version is obtained employing the Dirac-Impulse



Probability density function



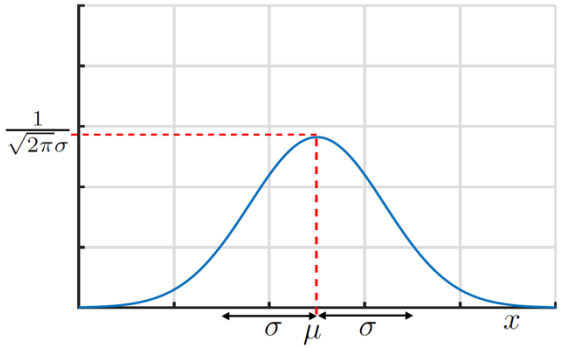
(Cumulative) distribution function

Gaussian-Distribution (Normal Distribution $X \sim N(\mu, \sigma^2)$):

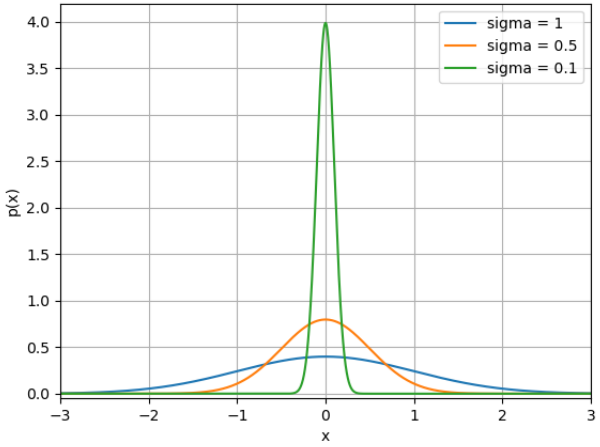
- The probability density function reads:

$$p_X(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

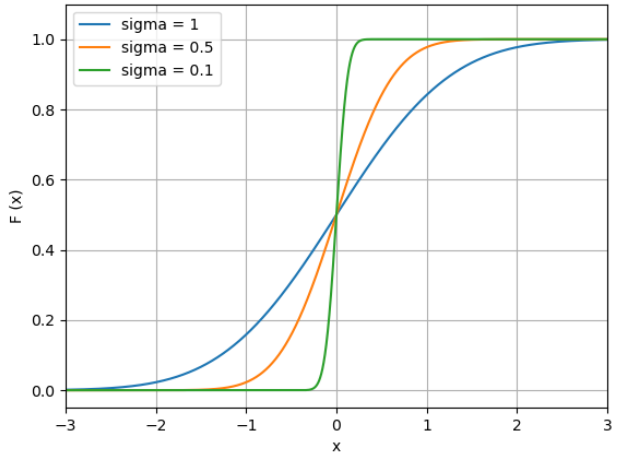
  - mean μ, standard deviation σ
- For one-dimensional distributions:
  - ~68.3% of all realizations of X are in the interval $[\mu - \sigma, \mu + \sigma]$
  - ~95.4% are in $[\mu - 2\sigma, \mu + 2\sigma]$
  - ~99.7% are in $[\mu - 3\sigma, \mu + 3\sigma]$



Gaussian/Normal Distribution



Probability density function $\mu = 0$



(Cumulative) distribution function, $\mu = 0$

Central Limit Theorem

Central Limit Theorem

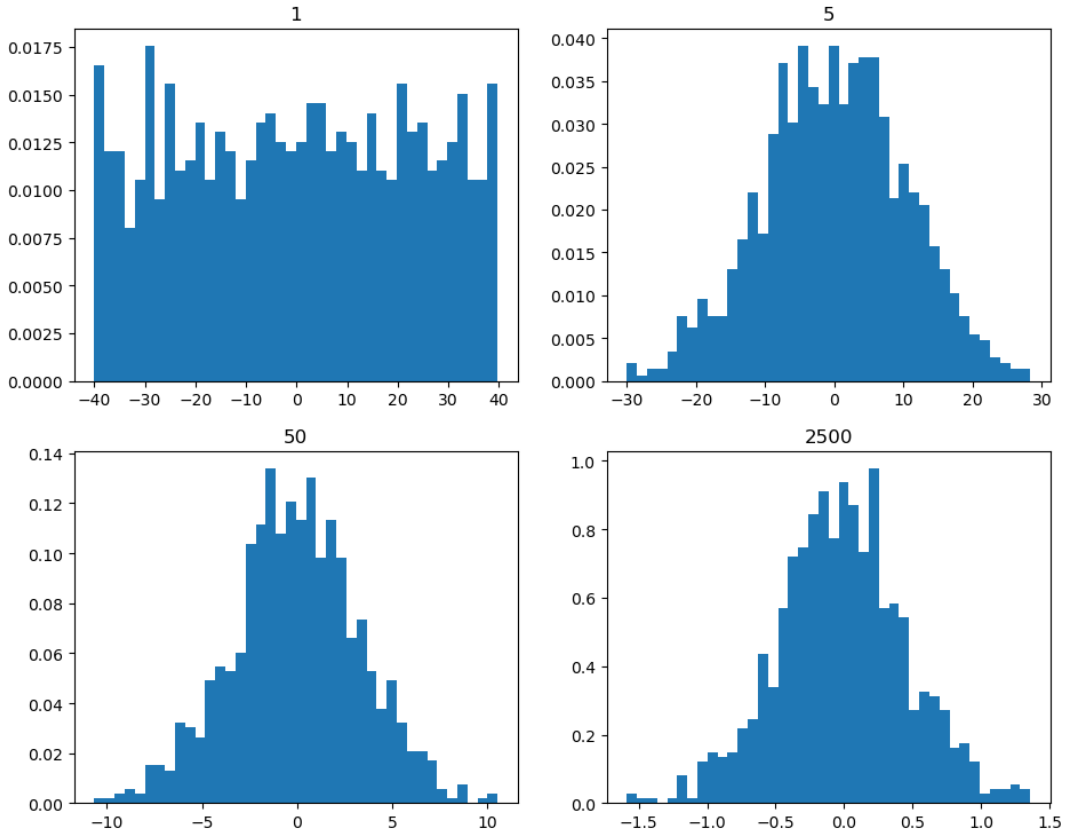
- The average of differently, stochastically independently distributed random variables approaches a Gaussian distribution with an increasing number of samples, i. e.. $X_1, X_2, \ldots, X_n$ are stochastically independent random variables, then

$$\overline{X} = \lim_{N \to \infty} \left(\frac{1}{N} \sum_{i=1}^{N} X_i \right)$$

  is normally distributed
- This is independent of the underlying distribution of X_i
- The central limit theorem is therefore the argument for the assumption of a Gaussian distribution in many different applications

Example: Central Limit Theorem

- We generate 4 sequences $X_1, X_2, \ldots X_N$ of uniformly distributed random variables in the interval $[-40, 40]$ with $N = 1, N = 5$, $N = 50$ and $N = 2500$ and compute the average
- This is done 1000 times and we compute the histogram of the average

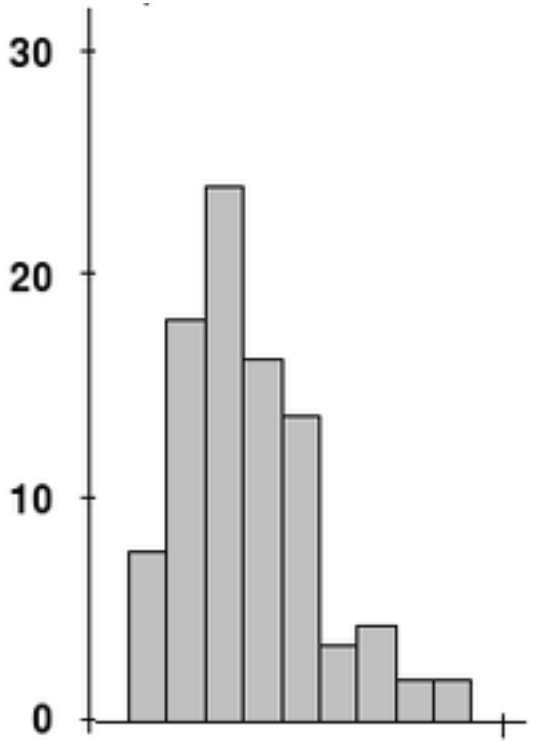


Histograms of the avarages of the sequances uniformly distributed random variables

Moments and Central Moments of Probability Distributions

- **Moments of a function: Quantitative measures related to the shape of the function's graph.**
  - If the function is a probability distribution, then
    - the first moment is the expected value,
    - the second central moment is the variance,
    - the third standardized moment is the skewness (0 for symmetric distributions).

Moment ordinal	Moment		
	Raw	Central	Standardized
1	Mean	0	0
2	—	Variance	1
3	—	—	Skewness



- **Moments are calculated**
  - **raw:** around zero (i.e. without regard for the mean of the distribution)
  - **central:** around the mean
  - **standardized:** around the mean and divided by the variance

Moments of a Probability Distribution

- The i-th **moment** (non-central, non-standardized) of a probability distribution is computed as

$$E\left[X^i\right] = \int_{-\infty}^{\infty} x^i \cdot p_X(x)dx \quad \text{or} \quad E\left[X^i\right] = \sum_{k\in\mathcal{N}} x(k)^i \cdot p_X(x(k))$$

- Example for our dice throw: $E[\text{dice}] = \frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6} = \frac{21}{6} = 3.5$

- The first moment is denoted as mean $\boldsymbol{\mu}$.

$$\mu \equiv E[X]$$

Moments of a Probability Distribution

- **Central moments** are calculated by subtracting the mean first.

  The i-th **central moment** of a probability distribution is

$$\begin{aligned} & E[(X - E[X])^i] \\ = & E[(X - \mu)^i] \\ = & \int_{-\infty}^{\infty} (x - \mu)^i \cdot p_X(x) dx \end{aligned}$$

  or $\sum_{k \in N} (x(k) - \mu)^i \cdot p_X(x(k))$ for the discrete case.

- The **second central moment is called variance** $\mathrm{Var}(X)$**,** and it square root is called the **standard deviation.**

$$\mathrm{Var}(X) = E\left[(X - E[X])^2\right] \quad \text{and} \quad \sigma = \sqrt{\mathrm{Var}[X]}$$

Computational Rules for the Mean

- For two random variables X and Y as well as constants a and b the following rules hold:

$$\begin{aligned} E[a] &= a, \\ E[aX] &= aE[X], \\ E[a+bX] &= a + bE[X], \\ E[X+Y] &= E[X] + E[Y] \end{aligned}$$

and for stochastically independent random variables X and Y

$$E[X \cdot Y] = E[X] \cdot E[Y]$$

holds.

Computational Rules for the Variance

- For two random variables X and Y as well as constants a and b the following rules hold:

$$\begin{aligned} \mathrm{Var}[a] &= 0, \\ \mathrm{Var}[a+X] &= \mathrm{Var}[X], \\ \mathrm{Var}[a+bX] &= b^2 \cdot \mathrm{Var}[X], \\ \mathrm{Var}[X+Y] &= \mathrm{Var}[X] + \mathrm{Var}[Y] \\ &\quad + 2 \cdot \underbrace{(E[X \cdot Y] - E[X]\,E[Y])}_{\mathrm{Cov}(X,Y)} \end{aligned}$$

and for statistically independent random variables X and Y

$$\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$$

holds.



Technische Hochschule Ingolstadt

Fakultät Elektro- und Informationstechnik

Thank you very much!